- There are 4 Questions for 30 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof).
- You do not need any calculators.
- Good luck!

1. [5] What is the maximum value of $x+2y+3z$ over all the points in the unit sphere (i.e. over all points (x,y,z) that satisfy $x^2+y^2+z^2=1$) ? Clearly justify your answer.

2. [5] For vectors $\underline{x}, \underline{y} \in \mathbb{R}^2$, consider the following definition

$$\langle \underline{x}, \underline{y} \rangle = 5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1.$$

   Is this a valid inner product ?

3. Consider the following sets in $\mathbb{R}^2$:

$$S_1 = \{(x_1,x_2) : x_1 = x_2\}, \quad S_2 = \{(x_1,x_2) : x_1^2 + x_2^2 = 1\}$$

   (a) [5] Find point on S_1 that is closest to $\underline{y} = (1,0)^\top$, i.e. find

$$\arg\min_{\underline{x}\in S_1} \|\underline{x} - \underline{y}\|_p$$

   for each $p = 1, 2, \infty$.

   (b) [5] Find the points on S_2 which have the smallest $p-$ norm, for $p = 1, 2, \infty$.

4. [10] Let A be an $m \times n$ real matrix.

   (a) Show that $A\underline{x} = 0$ if and only if $A^\top A\underline{x} = 0$.

   (b) What can you say about the relationship between nullity(A) and nullity$(A^\top A)$?

   (c) What is the relationship between rank(A) and rank$(A^\top A)$?

   (d) State true or false: $A^\top A$ is invertible if and only if A has linearly independent columns.

   Consider the following systems of equations

$$\text{System}_1 : \ A\underline{x} = \underline{b} \quad \text{System}_2 : \ A^\top A\underline{x} = A^\top \underline{b}$$

   State True or false for the following, with justification

   (e) System$_1$ has a solution if and only if System$_2$ does.

   (f) If a solution to System$_1$ exists, it is given by $\underline{x} = (A^\top A)^{-1}\underline{b}$.